

P2 — Frontend & Mobile Lead

RoadAssist Bharat · 86 days across 3 reviews Owns: React Native app · web, government, admin and fleet portals · design system · offline client · maps UI · accessibility · 8 languages. Backs up **P4**.

Your three standing rules 1. The **reference device is the truth**: Android 10, 2 GB RAM, throttled 3G. If it does not work there, it does not work. 2. Every screen ships with **four** states built — loading, empty, error, offline. A happy path alone is not done. 3. No hardcoded user-facing text, ever. Eight languages from the first commit.

REVIEW 1 — Foundation · Weeks 1–5 · 17 days

Week	Task	Days
1	Device and connectivity study — measure real 2G/3G latency on 3 highway corridors; catalogue the Indian device mix	2
1–2	20 user interviews (UX focus); competitor teardown; accessibility and literacy findings	2
2	6 personas + 4 journey maps : highway breakdown, city no-start, night accident, fleet truck failure	2
2	Information architecture, navigation map, component inventory	2
3	Low-fidelity wireframes for the 12 critical screens with all four states	3
3–4	Design system foundations: colour, type, spacing, motion, touch targets; app scaffolds	3
4	Local offline database schema and storage budget (≤ 120 MB, with eviction)	1
5	Auth screens: phone entry, OTP with SMS auto-read, profile setup, consent centre	2
5	Secure token storage (Keychain/Keystore), silent refresh with request queueing, biometric gate	—

Deliverables

Device matrix · connectivity report · 6 personas · 4 journey maps · information architecture · wireframes for 12 screens · design system foundations · auth screens · secure storage · clickable prototype.

Your Review 1 demo — 5 minutes

Hold up the **reference device physically**. Show a competitor's app struggling on it — that is the brief. Then walk the clickable prototype of the complete breakdown journey in Hindi. Finish by logging in over throttled 3G with SMS auto-read, then killing the app mid-refresh to show it recovering cleanly.

Passed when

Wireframes cover loading, empty, error and offline for all 12 screens · design system renders in all 8 languages with no truncation · tokens stored only in Keychain/Keystore, never in plain storage · 10 concurrent 401s trigger exactly **one** refresh.

Your design constraints (evidence-based, from Week 1)

Works one-handed, in rain, at night, on a highway shoulder · readable in direct sunlight (contrast $\geq 7:1$) · comprehensible at low literacy (icon + colour + voice, never text alone) · never a blank screen · never a silent failure · $\text{APK} \leq 28 \text{ MB}$ · $\leq 30 \text{ MB}$ data per month.

REVIEW 2 — Core Product · Weeks 6–12 · 36 days

Week	Task	Days
6–7	Design system package — 42 components, dark + light, 8 languages, motion system	6
7–8	Citizen app: onboarding, garage, request-help flow , diagnosis, matching, live tracking, payment, history	12
9	Mechanic app: job feed, accept, navigation, checklist, parts, invoice builder, earnings	4
9	Web PWA parity for the core flows	2

Week	Task	Days
10	AI confidence UI + user feedback capture that feeds P3's training loop	2
11–12	Offline client — local database, priority operation queue, sync engine, conflict resolution screen, background sync	10
12	(within) <code>useOfflineQuery</code> as the only data-access pattern in the app — this is what makes offline uniform rather than per-screen	—

Note: Admin console and Fleet dashboard moved to P4 to keep this review achievable. Android Auto is the declared first cut if you slip.

Deliverables

Design system (42 components) · citizen app (Android + iOS) · mechanic app · web PWA · offline client library · sync status and conflict UI · Storybook · visual regression suite.

Your Review 2 demo — 8 minutes

The full journey on a real **₹8,000 Android phone**, throttled to 3G, in Hindi, one-handed, at maximum brightness to simulate sunlight. **No simulator**. Then switch to airplane mode: add a vehicle, run an on-device diagnosis, create a help request, view an offline map, cancel and re-create. Reconnect — everything syncs and one deliberate conflict is resolved in a screen that makes the choice obvious.

Passed when

Full journey works on Android 10 / 2 GB / 3G · every screen has all four states · airplane-mode journey syncs with zero data loss · cold start under 2.5 seconds on the reference device · optimistic writes visibly marked "pending" until confirmed — never a silent revert.

REVIEW 3 — Complete System · Weeks 13–18 · 33 days

Week	Task	Days
13	Maps UI: MapLibre components, offline map packs with resumable download, live tracking, landmark location picker	7
13	(within) Diagnosis UI: OBD Bluetooth pairing, guided photo capture, drivability verdict	—
14	SOS UI — long-press button, full-screen crash countdown over the lock screen, incident room	4
14–15	Government portal — live incident map (10,000 markers, no jank), blackspot analysis, 6 statutory report generators	6
15	Analytics dashboards; fleet and mechanic earnings screens	4
16	End-to-end tests (Detox + Playwright) for 12 journeys; visual regression baseline	4
16	Device farm across 12 Android + 4 iOS; accessibility audit; 8-language QA; bug fixes	3
17	App store submission with staged rollout; over-the-air update channel	3
17–18	Optimisation — APK size, startup, jank at 60fps, memory leaks, data usage	2

Deliverables

Maps components + offline packs · diagnosis flow · SOS and incident room · government portal · analytics dashboards · end-to-end test suites · device compatibility matrix · accessibility report · published apps · performance report.

Your Review 3 demo — 7 minutes

Download a district map pack over WiFi, go to airplane mode, search a landmark and get directions — fully offline. Then trigger a simulated crash **with the phone locked and in a pocket**: the countdown takes over the screen with alarm and vibration. Then the government portal: find a blackspot, generate the statutory PDF. Close with the killer demo — **the entire booking journey driven by a screen reader with the display switched off**. If a blind user can call for roadside assistance, the app is well built.

Passed when

Offline navigation works with the device in airplane mode · countdown renders over the lock screen and in Do Not Disturb · government map handles 10,000 incidents without jank · WCAG 2.1 AA passes a real screen-reader walkthrough (not just an automated scan) · APK ≤ 28 MB · cold start ≤ 2.0 seconds · ≤ 30 MB data per month.

Your Key Design Decisions (defend these to the panel)

Decision	Why
One shared component library across 5 apps	It is the only way one person ships five clients in 18 weeks.
Three portals, one Next.js app	Government, admin and fleet share a codebase with role-based routing instead of being three projects.
System fonts, no custom font download	Saves ~400 KB and renders all 8 Indian scripts correctly. Custom fonts routinely break Devanagari and Tamil.
FlashList everywhere, never FlatList	Long lists are where low-end Android devices actually fall over.
<code>useOfflineQuery</code> as the single data pattern	Offline becomes a property of the architecture rather than something re-implemented per screen.
Confidence shown as words, not percentages	"We're fairly sure" versus "this is a guess — a mechanic will confirm." Users cannot calibrate a 0.73.
Animation auto-disabled on low-end devices	Detected at runtime. Motion is a luxury; responsiveness is not.